Solution to MP363 Repeat Exam, Autumn 2015

(1)

P.1

(a) As stated, we need to find the lowest nonvanishing term in the Taylor series expansion of $B(\nu,T)$ around $\nu=0$. This term will give the small-ν behaviour of $B(\nu,T)$, because all terms with higher powers of ν will be negligible compared to it for values of ν near zero (or, more precisely, when $\nu \ll \frac{k_BT}{h}$). So we'll need to find the lowest derivative of $B(\nu,T)$ with respect to ν which is nonzero at $\nu=0$; this will require several uses of l'Hôpital's rule, as we'll see:

$$B(0,T) = \lim_{\nu\to0} \frac{2h}{c^2}\frac{\nu^3}{e^{h\nu/k_BT}-1} = \frac{2h}{c^2}\lim_{\nu\to0}\frac{3\nu^2}{\frac{h}{k_BT}e^{h\nu/k_BT}} = \frac{6k_BT}{c^2}\frac{0}{1}$$

$$=0,$$

so the first term in the Taylor series is zero. Now,

$$\frac{\partial B}{\partial\nu}(\nu,T) = \frac{2h}{c^2}\left[\frac{3\nu^2}{e^{h\nu/k_BT}-1} - \frac{h}{k_BT}\frac{\nu^3 e^{h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^2}\right], \text{ so}$$

$$\frac{\partial B}{\partial\nu}(0,T) = \frac{6h}{c^2}\lim_{\nu\to0}\left(\frac{\nu^2}{e^{h\nu/k_BT}-1}\right) - \frac{2h^2}{c^2k_BT}\lim_{\nu\to0}\left[\frac{\nu^3 e^{h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^2}\right]$$

$$= \frac{6h}{c^2}\lim_{\nu\to0}\frac{2\nu}{\frac{h}{k_BT}e^{h\nu/k_BT}} - \frac{2h^2}{c^2k_BT}\lim_{\nu\to0}\frac{3\nu^2e^{h\nu/k_BT}+\frac{h\nu^3}{k_BT}e^{\frac{h\nu}{k_BT}}}{\frac{2h}{k_BT}(e^{h\nu/k_BT}-1)e^{h\nu/k_BT}}$$

$$=0-\frac{h}{c^2}\lim_{\nu\to0}\frac{3\nu^2+\frac{h\nu^3}{k_BT}}{e^{h\nu/k_BT}-1}$$

$$=-\frac{h}{c^2}\lim_{\nu\to0}\frac{6\nu+\frac{3h\nu^2}{k_BT}}{\frac{h}{k_BT}e^{h\nu/k_BT}}=0$$

so this term in the Taylor series vanishes as well. Next, $\frac{\partial^2B(\nu,T)}{\partial\nu^2}$:

$$\frac{\partial^2B}{\partial\nu^2}(\nu,T) = \frac{\partial}{\partial\nu}\left[\frac{2h}{c^2}\left(\frac{3\nu^2}{e^{h\nu/k_BT}-1}-\frac{h}{k_BT}\frac{\nu^3e^{h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^2}\right)\right]$$

$$=\frac{2h}{c^2}\left[\frac{6\nu}{e^{h\nu/k_BT}-1}-\frac{6h}{k_BT}\frac{\nu^2e^{h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^2}+\frac{2h^2}{k_B^2T^2}\frac{\nu^3e^{2h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^3}\right]$$

Look at each limit separately:

$$\lim_{\nu\to0}\frac{\nu}{e^{h\nu/k_BT}-1} = \lim_{\nu\to0}\frac{1}{\frac{h}{k_BT}e^{\frac{h\nu}{k_BT}}} = \frac{k_BT}{h}$$

$$\lim_{\nu\to0}\frac{\nu^2e^{h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^2} = \lim_{\nu\to0}\frac{2\nu e^{h\nu/k_BT}+\frac{h\nu^2}{k_BT}e^{\frac{h\nu}{k_BT}}}{2\frac{h}{k_BT}e^{\frac{h\nu}{k_BT}}(e^{\frac{h\nu}{k_BT}}-1)} = \lim_{\nu\to0}\frac{2\nu+\frac{h\nu^2}{k_BT}}{\frac{2h}{k_BT}(e^{\frac{h\nu}{k_BT}}-1)}$$

$$=\frac{k_BT}{2h}\lim_{\nu\to0}\frac{2+\frac{2h\nu}{k_BT}}{\frac{h}{k_BT}e^{\frac{h\nu}{k_BT}}} = \frac{k_B^2T^2}{h^2}$$

$$\lim_{\nu\to 0} \frac{\nu^3 e^{2h\nu/k_BT}}{(e^{h\nu/k_BT}-1)^3} = \lim_{\nu\to 0} \frac{3\nu^2 e^{\frac{2h\nu}{k_BT}} + \frac{2h}{k_BT}\nu^3 e^{\frac{2h\nu}{k_BT}}}{\frac{3h}{k_BT} e^{\frac{h\nu}{k_BT}}(e^{\frac{h\nu}{k_BT}}-1)^2}$$

$$= \frac{k_BT}{3h}\lim_{\nu\to 0} \frac{\left(3\nu^2 + \frac{2h\nu^3}{k_BT}\right) e^{\frac{h\nu}{k_BT}}}{(e^{\frac{h\nu}{k_BT}}-1)^2}$$

$$= \frac{k_BT}{3h}\lim_{\nu\to 0} \frac{\left(6\nu + \frac{6h\nu^2}{k_BT}\right)e^{\frac{h\nu}{k_BT}} + \frac{h}{k_BT}\left(3\nu^2+\frac{2h\nu^3}{k_BT}\right)e^{\frac{h\nu}{k_BT}}}{\frac{2h}{k_BT}e^{h\nu/k_BT}(e^{\frac{h\nu}{k_BT}}-1)}$$

$$= \frac{k_B^2T^2}{6h^2}\lim_{\nu\to 0} \left[\frac{6\nu + \frac{9h\nu^2}{k_BT} + \frac{2h^2\nu^3}{k_B^2T^2}}{e^{h\nu/k_BT}-1}\right]$$

$$= \frac{k_B^2T^2}{6h^2}\lim_{\nu\to 0} \left|\frac{6+\frac{18h\nu}{k_BT}+\frac{6h^2\nu^2}{k_B^2T^2}}{\frac{h}{k_BT}e^{h\nu/k_BT}}\right| = \frac{k_B^3T^3}{h^3}.$$

Thus,

$$\frac{\partial^2 B}{\partial\nu^2}(0,T) = \frac{2h}{c^2}\left[6\cdot\left(\frac{k_BT}{h}\right) - \frac{6h}{k_BT}\left(\frac{k_B^2T^2}{h^2}\right) + \frac{2h^2}{k_B^2T^2}\left(\frac{k_B^3T^3}{h^3}\right)\right]$$

$$= \frac{4k_BT}{c^2}$$

Thus, since

$$B(\nu,T) = B(0,T) + \frac{\partial B}{\partial\nu}(0,T)\,\nu + \frac{1}{2}\frac{\partial^2B}{\partial\nu^2}(0,T)\,\nu^2 + \frac{1}{6}\frac{\partial^3B}{\partial\nu^3}(0,T)\,\nu^3 + \dots$$

we see that

$$B(\nu,T) = 0 + 0 + \frac{1}{2}\left(\frac{4k_BT}{c^2}\right)\nu^2 + O(\nu^3)$$
$$= \frac{2k_BT\nu^2}{c^2} + O(\nu^3) = \boxed{B_{RJ}(\nu,T)+O(\nu^3)}$$ [30 pts]

So Planck's formula does indeed agree (approximately) when $\nu \ll k_BT/h$.

(b) This is quite straightforward: we're given $\frac{d\Phi}{d\nu}$, so the total flux emitted by all frequencies is simply the integral of this from $\nu=0$ to $\nu\to\infty$, i.e. $\Phi = \int_0^\infty \frac{d\Phi}{d\nu}\,d\nu$. But $\frac{d\Phi}{d\nu}$ is given in terms of $B(\nu,T)$, so

$$\Phi = \int_0^\infty \pi B(\nu,T)\,d\nu = \frac{2\pi h}{c^2}\int_0^\infty \frac{\nu^3}{e^{h\nu/k_BT}-1}\,d\nu.$$

Let $u = h\nu/k_BT$ be our change of variable, so that

$$\Phi = \frac{2\pi h}{c^2}\int_0^\infty \frac{(k_BTu/h)^3\left(\frac{k_BT}{h}du\right)}{e^u-1} = \frac{2\pi k_B^4T^4}{h^3c^2}\int_0^\infty\frac{u^3\,du}{e^u-1}.$$

The integral is given to be $\pi^4/15$, so we see

$$\Phi = \frac{2\pi^5k_B^4}{15h^3c^2}T^4 = \boxed{\sigma T^4}$$ [10 pts]

where $\boxed{\sigma = \frac{2\pi^5k_B^4}{15h^3c^2}}$ is the Stefan-Boltzmann constant. [10 pts]

P.2

(a) One of the defining property of an inner product is that, for any two vectors ψ and ζ, $\langle\psi|\zeta\rangle = \langle\zeta|\psi\rangle^*$. Furthermore, [5pts] if A is an operator, then $A^\dagger$ is the operator that satisfies $\langle\psi|A\zeta\rangle = \langle A^\dagger\psi|\zeta\rangle$. However, the right-hand side of [5pts] this eqn is also $\langle\zeta|A^\dagger\psi\rangle^*$, so

$$\boxed{\langle\psi|A\zeta\rangle = \langle\zeta|A^\dagger\psi\rangle^*}$$ [5pts]

as desired.

(b) The easiest way to do this is to show that $\langle\psi|a\zeta\rangle - \langle\zeta|a^\dagger\psi\rangle^*$ vanishes for any two states ψ and ζ; this will prove that $a^\dagger$ is the adjoint of a. We see that

$$\langle\psi|a\zeta\rangle - \langle\zeta|a^\dagger\psi\rangle^* = \int_{-\infty}^{\infty}\psi^*(a\zeta)\,dx - \left(\int_{-\infty}^{\infty}\zeta^*(a^\dagger\psi)\,dx\right)^*$$

$$= \int_{-\infty}^{\infty}\psi^*\left[\sqrt{\frac{m\omega}{2\hbar}}\left(x\zeta + \frac{\hbar}{m\omega}\frac{\partial\zeta}{\partial x}\right)\right]dx - \int_{-\infty}^{\infty}\left[\sqrt{\frac{m\omega}{2\hbar}}\left(x\psi - \frac{\hbar}{m\omega}\frac{\partial\psi}{\partial x}\right)\right]^*\zeta\,dx$$

$$= \sqrt{\frac{m\omega}{2\hbar}}\int_{-\infty}^{\infty}\left[\left(\psi^* x\zeta + \frac{\hbar}{m\omega}\psi^*\frac{\partial\zeta}{\partial x}\right) - \left(x\psi^*\zeta - \frac{\hbar}{m\omega}\frac{\partial\psi^*}{\partial x}\zeta\right)\right]dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\int_{-\infty}^{\infty}\left(\psi^*\frac{\partial\zeta}{\partial x} + \frac{\partial\psi^*}{\partial x}\zeta\right)dx$$

since $\psi^* x\zeta$ and $x\psi^*\zeta$ cancel out. The quantity in the brackets is just $\frac{\partial}{\partial x}(\psi^*\zeta)$, so

$$\langle\psi|a\zeta\rangle - \langle\zeta|a^\dagger\psi\rangle^* = \sqrt{\frac{\hbar}{2m\omega}}\int_{-\infty}^{\infty}\frac{\partial}{\partial x}(\psi^*\zeta)\,dx$$

$$= \sqrt{\frac{\hbar}{2m\omega}}\left|\psi^*(x)\zeta(x)\right|_{-\infty}^{\infty}$$

by the Fundamental Theorem of Calculus. Now, all wavefunctions must be normalized such that $\int_{-\infty}^{\infty}|\psi(x)|^2dx = 1$, which requires that $\boxed{\psi(x) \text{ must vanish as } x\to\pm\infty}$. Otherwise $\int_{-\infty}^{\infty}|\psi|^2dx$ would [15pts] be undefined. Therefore, this condition says that $\psi^*\zeta$ must be zero at both $x\to\infty$ and $x\to-\infty$, and so

$$\langle\psi|a\zeta\rangle - \langle\zeta|a\psi\rangle^* = 0$$

and thus $\boxed{a^\dagger \text{ is the adjoint of } a.}$ [20pts]

P.3

(a) When the wavefunction of a state ψ is written in terms of energy eigenstates $\{\phi_n\}$ as $\psi = \sum_n c_n \phi_n$, then $|c_n|^2$ is the probability that a measurement of energy will give E_n. Here, since $\psi = \frac{1}{\sqrt{2}}\phi_1 - \frac{1}{\sqrt{2}}\phi_3$, we see immediately that $c_1 = \frac{1}{\sqrt{2}}$, $c_3 = -\frac{1}{\sqrt{2}}$ and $c_n = 0$ for $n \neq 1, 3$. Therefore, we have that $|c_n|^2 = 0$ unless $n = 1$ or 3, and thus the only possible results of measuring energy will be

$$\boxed{E_1 = \frac{\pi^2\hbar^2}{2ma^2} \quad \text{and} \quad E_3 = \frac{9\pi^2\hbar^2}{2ma^2}}$$ [5 pts each]

with probabilities

$$\boxed{|c_1|^2 = \frac{1}{2} = 50\% \quad \text{and} \quad |c_3|^2 = \frac{1}{2} = 50\%}$$ [5 pts each]

(b) Let's find $\langle p \rangle$ first, because it's the easiest: by defn,

$$\langle p \rangle = \langle \psi | p \psi \rangle$$

$$= \int_{-\infty}^{\infty} \psi^*(t,x)\left(p\psi(t,x)\right)dx = -i\hbar \int_{-\infty}^{\infty} \psi^*(t,x) \frac{\partial \psi(t,x)}{\partial x} dx.$$

Now,

$$\frac{\partial \psi(t,x)}{\partial x} = \frac{1}{\sqrt{2}} u_1'(x) e^{-iE_1 t/\hbar} - \frac{1}{\sqrt{2}} u_3'(x) e^{-iE_3 t/\hbar}$$

$$= \frac{1}{\sqrt{a}}\left[\frac{\pi}{a}\cos\left(\frac{\pi x}{a}\right)e^{-iE_1 t/\hbar} - \frac{3\pi}{a}\cos\left(\frac{3\pi x}{a}\right)e^{-iE_3 t/\hbar}\right]$$

so

$$\langle p \rangle = -\frac{i\hbar\pi}{a^2}\int_0^a \left[\sin\left(\frac{\pi x}{a}\right)e^{-iE_1 t/\hbar} - \sin\left(\frac{3\pi x}{a}\right)e^{-iE_3 t/\hbar}\right]^* \left[\cos\left(\frac{\pi x}{a}\right)e^{-iE_1 t/\hbar} - 3\cos\left(\frac{3\pi x}{a}\right)e^{-iE_3 t/\hbar}\right]dx$$

$$= -\frac{i\hbar\pi}{a^2}\int_0^a \Big[\sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{\pi x}{a}\right) + 3\sin\left(\frac{3\pi x}{a}\right)\cos\left(\frac{3\pi x}{a}\right) - \sin\left(\frac{3\pi x}{a}\right)\cos\left(\frac{\pi x}{a}\right)e^{i(E_3-E_1)t/\hbar}$$
$$- 3\sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{3\pi x}{a}\right)e^{i(E_1-E_3)t/\hbar}\Big]dx$$

Note that, if $m \neq n$,

$$\int_0^a \sin\left(\frac{m\pi x}{a}\right)\cos\left(\frac{n\pi x}{a}\right)dx = \frac{1}{2}\int_0^a \left[\sin\left(\frac{(m+n)\pi x}{a}\right) + \sin\left(\frac{(m-n)\pi x}{a}\right)\right]dx$$

$$= \frac{1}{2}\left[\frac{-a}{\pi(m+n)}\cos\left(\frac{(m+n)\pi x}{a}\right) - \frac{a}{\pi(m-n)}\cos\left(\frac{(m-n)\pi x}{a}\right)\right]\Bigg|_0^a$$

$$= \frac{-a}{2\pi}\left[\frac{(-1)^{m+n}-1}{m+n} + \frac{(-1)^{m-n}-1}{m-n}\right]$$

The last two terms in the integrand have $m=3, n=1$ and $m=1, n=3$ respectively, i.e. both $m+n$ and $m-n$ are even. Thus, $(-1)^{m+n} = (-1)^{m-n} = 1$, and so those two integrals vanish, and so

$$\langle P\rangle = -\frac{i\hbar\pi}{a^2}\int_0^a \left[\sin\left(\frac{\pi x}{a}\right)\cos\left(\frac{\pi x}{a}\right) + 3\sin\left(\frac{3\pi x}{a}\right)\cos\left(\frac{3\pi x}{a}\right)\right]dx$$

$$= -\frac{i\hbar\pi}{2a^2}\int_0^a \left[\sin\left(\frac{2\pi x}{a}\right) + 3\sin\left(\frac{6\pi x}{a}\right)\right]dx$$

$$= -\frac{i\hbar\pi}{2a^2}\left[-\frac{a}{2\pi}\cos\left(\frac{2\pi x}{a}\right) - \frac{a}{2\pi}\cos\left(\frac{6\pi x}{a}\right)\right]_0^a = 0$$

because $\cos(2\pi) = \cos(6\pi) = \cos 0 = 1$. Thus, $\boxed{\langle P\rangle = 0.}$ [10 pts]

Now for $\langle X\rangle$: here we get

$$\langle X\rangle = \int_{-\infty}^{\infty} \psi^*(t,x)\big(X\psi(t,x)\big)dx = \int_{-\infty}^{\infty} x\,|\psi(t,x)|^2\,dx$$

$$= \frac{1}{a}\int_0^a x\left|\sin\left(\frac{\pi x}{a}\right)e^{-\frac{iE_1 t}{\hbar}} - \sin\left(\frac{3\pi x}{a}\right)e^{-\frac{iE_3 t}{\hbar}}\right|^2 dx$$

$$= \frac{1}{a}\int_0^a x\left[\sin^2\left(\frac{\pi x}{a}\right) + \sin^2\left(\frac{3\pi x}{a}\right) - 2\sin\left(\frac{\pi x}{a}\right)\sin\left(\frac{3\pi x}{a}\right)\cos\left(\frac{(E_3-E_1)t}{\hbar}\right)\right]dx.$$

$$= \frac{1}{2a}\int_0^a x\left[\left(1-\cos\left(\frac{2\pi x}{a}\right)\right) + \left(1-\cos\left(\frac{6\pi x}{a}\right)\right) - 2\left[\cos\left(\frac{2\pi x}{a}\right) - \cos\left(\frac{4\pi x}{a}\right)\right]\cos\left(\frac{(E_3-E_1)t}{\hbar}\right)\right]dx$$

Note that

$$\int_0^a x\cos\left(\frac{2n\pi x}{a}\right)dx = \left[\frac{a}{2n\pi}x\sin\left(\frac{2n\pi x}{a}\right) + \frac{a^2}{4n^2\pi^2}\cos\left(\frac{2n\pi x}{a}\right)\right]_0^a$$

$$= \frac{a}{2n\pi}\left|a\sin(2n\pi)\right| + \frac{a^2}{4n^2\pi^2}\cos(2n\pi) - 0 - \frac{a^2}{4n^2\pi^2}\cos 0\Big|$$

$$= 0$$

for any positive integer n. Since all of the cosine terms in the integrand have this form, they all vanish, leaving

$$\langle X\rangle = \frac{1}{2a}\int_0^a 2x\,dx = \frac{1}{2a}x^2\Big|_0^a = \boxed{a/2}.$$

[15 pts]

(c) This is trivial: we've just shown $\langle X\rangle$ = constant, so $\frac{d}{dt}\langle X\rangle = 0$. But $\langle P\rangle = 0$, so our results are consistent with the general result $\boxed{\langle P\rangle = m\frac{d}{dt}\langle X\rangle.}$ [5pts]